Name :- Mihir Mungara

Airflow Assignment – 3

DAG Code :-

===

```
from airflow import DAG

from airflow.sdk import Variable

from datetime import datetime, timedelta

from airflow.providers.amazon.aws.transfers.s3_to_sql import S3ToSqlOperator

from airflow.providers.common.sql.operators.sql import SQLExecuteQueryOperator



# ── Airflow Variables ──────────────────────────────────

S3_BUCKET = Variable.get("s3_bucket_name")

S3_FILE_KEY = Variable.get("s3_file_key")

TARGET_TABLE = Variable.get("target_table")



# ── Default Arguments ──────────────────────────────────

default_args = {

    "owner": "airflow",

    "depends_on_past": False,

    "email_on_failure": False,

    "retries": 2,

    "retry_delay": timedelta(minutes=5),

}


# ── DAG Definition ──────────────────────────────────

with DAG(

    dag_id="blob_storage_to_database",

    default_args=default_args,

    description="Load CSV file from AWS S3 into PostgreSQL",

    schedule="10 12 * * *",

    start_date=datetime(2024, 1, 1),

    catchup=False,
```

```
    tags=["s3", "postgres", "etl"],
) as dag:

    # ── Task 1: Create Table ──────────────────────────
    create_table = SQLExecuteQueryOperator(
        task_id="create_table_if_not_exists",
        conn_id="postgres_db_conn",
        sql=f"""
        CREATE TABLE IF NOT EXISTS {TARGET_TABLE} (
            id SERIAL PRIMARY KEY,
            order_id TEXT,
            customer_id TEXT,
            region TEXT,
            product TEXT,
            order_amount NUMERIC,
            order_date DATE,
            order_timestamp TIMESTAMP,
            loaded_at TIMESTAMP DEFAULT NOW()
        );
        """,
    )

    # ── Task 2: Load S3 CSV Into PostgreSQL ────────────
    load_s3_to_db = S3ToSqlOperator(
        task_id="load_s3_file_to_postgres",
        aws_conn_id="aws_s3_conn",
        sql_conn_id="postgres_db_conn",
        s3_bucket=S3_BUCKET,
        s3_key=S3_FILE_KEY,
        table=TARGET_TABLE,

        column_list=[
            "order_id",
            "customer_id",
            "region",
            "product",
            "order_amount",
```

```
        "order_date",
        "order_timestamp",
    ],

    commit_every=1000,

    parser=lambda filename: list(
        __import__("csv").reader(open(filename))
    )[1:],   # Skip header row
)

# ── Task 3: Verify Load ──────────────────────────────
verify_load = SQLExecuteQueryOperator(
    task_id="verify_row_count",
    conn_id="postgres_db_conn",
    sql=f"""
    SELECT COUNT(*) AS total_rows
    FROM {TARGET_TABLE};
    """,
)

# ── Task Dependencies ──────────────────────────────
create_table >> load_s3_to_db >> verify_load
```

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Connections :-

2 Connections

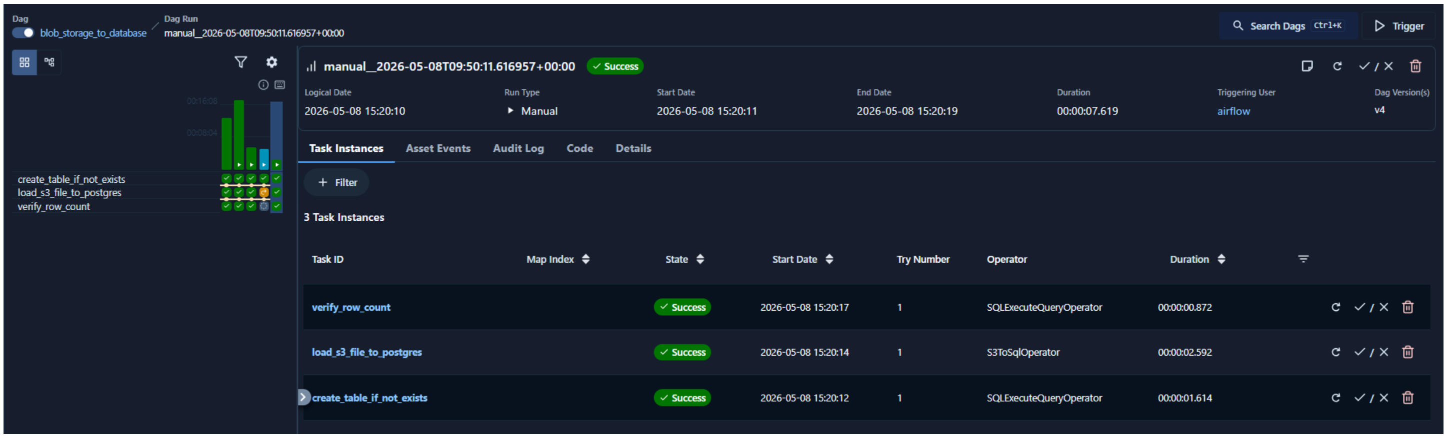
Connection ID	Connection Type	Description	Host	Port
aws_s3_conn	aws			
postgres_db_conn	postgres		postgres	5432

Variables :-

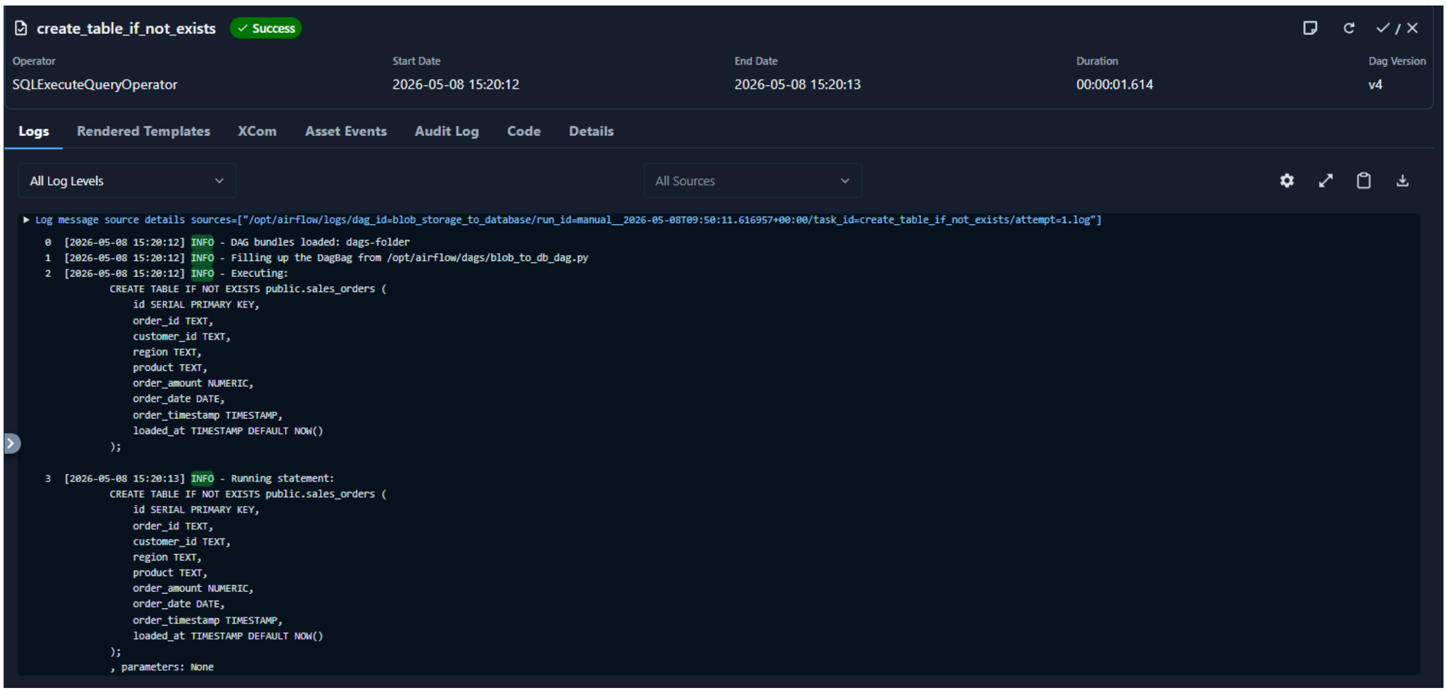
3 Variables

Key	Value	Description	Is Encrypted
s3_bucket_name	my-data-bucket-mm		true
s3_file_key	data/input/sales_data.csv		true
target_table	public.sales_orders		true

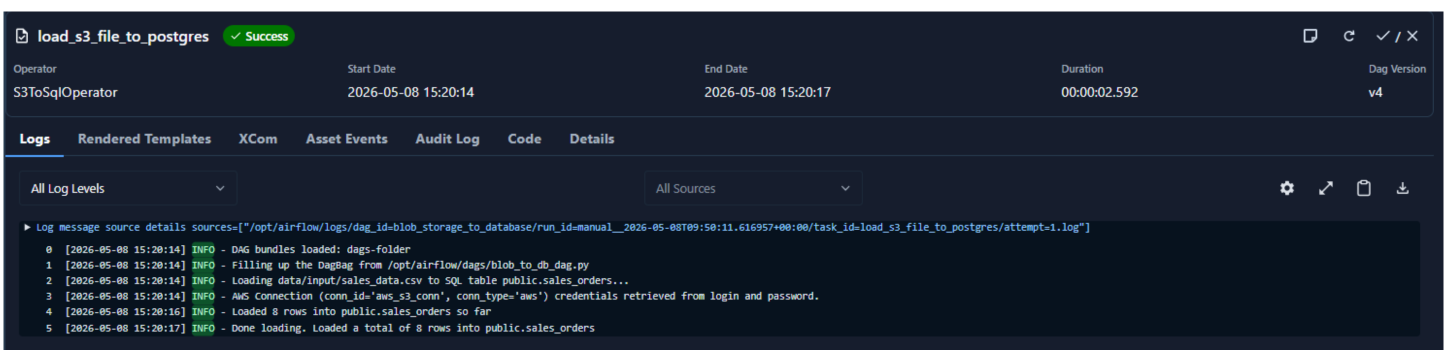
DAG Run Details :-



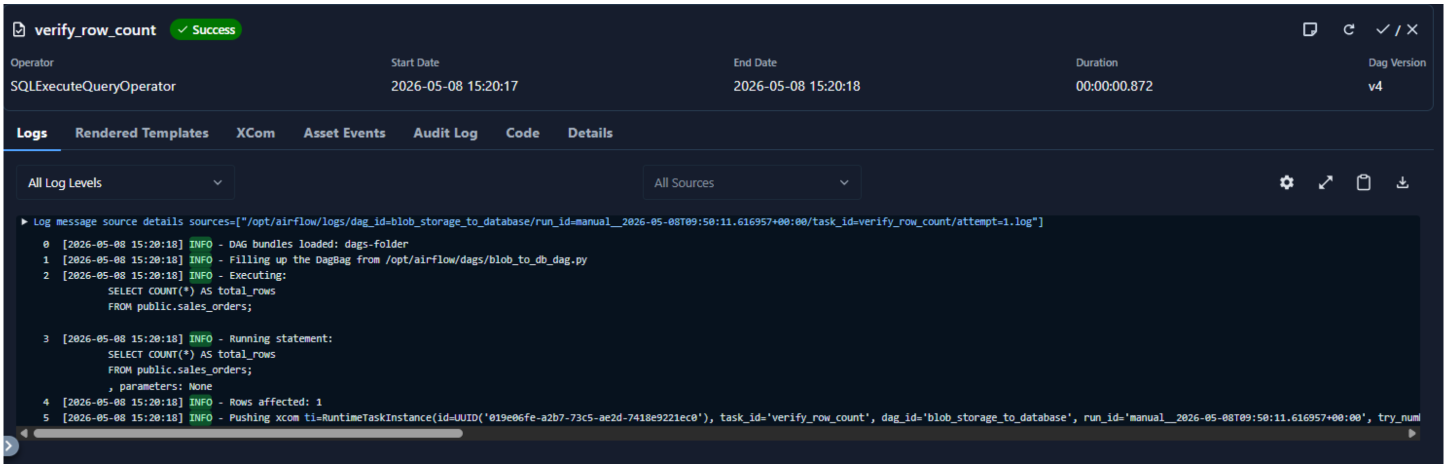
Logs of Task-1 : Creating Table if not Exists



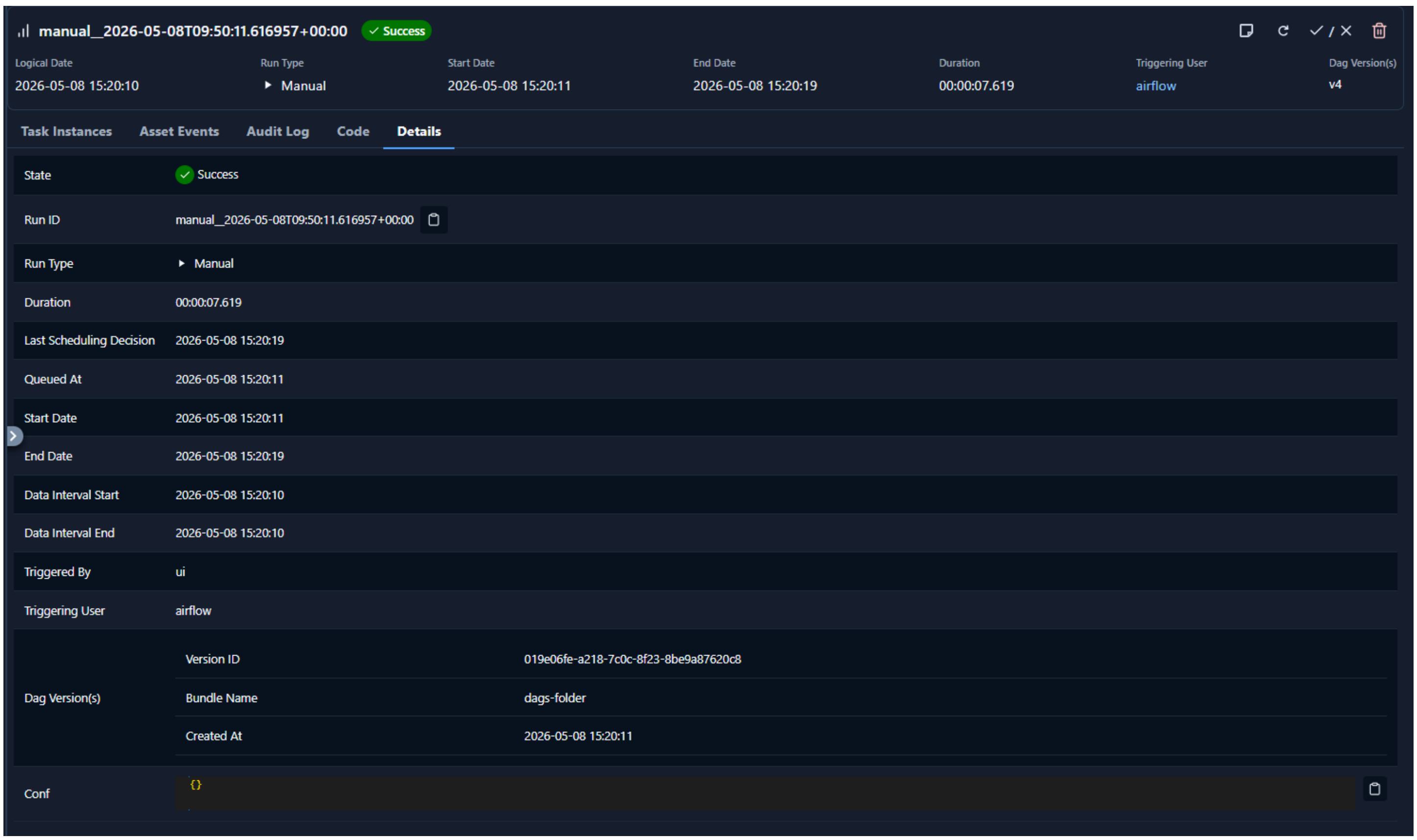
Logs of Task-2 : Loading S3 file to postgres



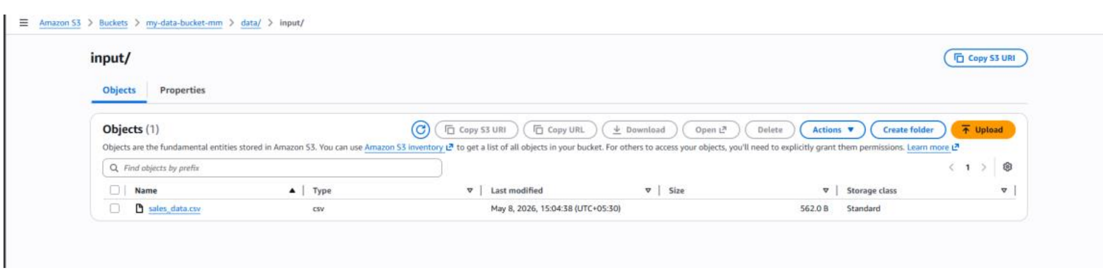
Logs of Task-3 : Verifying Row Count



Detailed DAG Run Details :-



S3 Bucket Configuration :-



Verfiying by Loading Data from Table

```
PS C:\Users\mihir.mungara\Desktop\Airflow_Practice> docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED          STATUS                    PORTS      NAMES
5d434364de8e   apache/airflow:3.2.1   "/bin/bash -c 'if [[…"   47 minutes ago   Up 27 seconds             8080/tcp   airflow_practice-airflow-init-1
80a9490172ad   postgres:13            "docker-entrypoint.s…"   47 minutes ago   Up 33 seconds (healthy)   5432/tcp   airflow_practice-postgres-1
```

```
PS C:\Users\mihir.mungara\Desktop\Airflow_Practice> docker exec -it airflow_practice-postgres-1 psql -U airflow -d airflow
psql (13.23 (Debian 13.23-1.pgdg13+1))
Type "help" for help.

airflow=# \dt
airflow=# SELECT * FROM public.sales_orders;
airflow=# SELECT * FROM public.sales_orders;
 id | order_id | customer_id | region | product | order_amount | order_date |   order_timestamp   |         loaded_at
----+----------+-------------+--------+---------+--------------+------------+---------------------+----------------------------
  1 | 1001     | C001        | North  | Laptop  |        75000 | 2023-01-10 | 2023-01-10 10:15:00 | 2026-05-08 09:50:16.676772
  2 | 1002     | C002        | South  | Mobile  |        30000 | 2023-01-12 | 2023-01-12 11:20:00 | 2026-05-08 09:50:16.676772
  3 | 1003     | C001        | North  | Tablet  |        20000 | 2023-02-05 | 2023-02-05 09:10:00 | 2026-05-08 09:50:16.676772
  4 | 1004     | C003        | East   | Laptop  |        80000 | 2023-02-20 | 2023-02-20 14:45:00 | 2026-05-08 09:50:16.676772
  5 | 1005     | C004        | West   | Mobile  |        28000 | 2023-03-01 | 2023-03-01 16:30:00 | 2026-05-08 09:50:16.676772
  6 | 1006     | C002        | South  | Laptop  |        72000 | 2023-03-15 | 2023-03-15 18:25:00 | 2026-05-08 09:50:16.676772
  7 | 1007     | C001        | North  | Mobile  |        32000 | 2023-03-18 | 2023-03-18 20:10:00 | 2026-05-08 09:50:16.676772
  8 | 1008     | C003        | East   | Tablet  |        22000 | 2023-04-02 | 2023-04-02 12:00:00 | 2026-05-08 09:50:16.676772
  9 | 1001     | C001        | North  | Laptop  |        75000 | 2023-01-10 | 2023-01-10 10:15:00 | 2026-05-08 09:51:14.595675
 10 | 1002     | C002        | South  | Mobile  |        30000 | 2023-01-12 | 2023-01-12 11:20:00 | 2026-05-08 09:51:14.595675
 11 | 1003     | C001        | North  | Tablet  |        20000 | 2023-02-05 | 2023-02-05 09:10:00 | 2026-05-08 09:51:14.595675
 12 | 1004     | C003        | East   | Laptop  |        80000 | 2023-02-20 | 2023-02-20 14:45:00 | 2026-05-08 09:51:14.595675
 13 | 1005     | C004        | West   | Mobile  |        28000 | 2023-03-01 | 2023-03-01 16:30:00 | 2026-05-08 09:51:14.595675
 14 | 1006     | C002        | South  | Laptop  |        72000 | 2023-03-15 | 2023-03-15 18:25:00 | 2026-05-08 09:51:14.595675
 15 | 1007     | C001        | North  | Mobile  |        32000 | 2023-03-18 | 2023-03-18 20:10:00 | 2026-05-08 09:51:14.595675
 16 | 1008     | C003        | East   | Tablet  |        22000 | 2023-04-02 | 2023-04-02 12:00:00 | 2026-05-08 09:51:14.595675
 17 | 1001     | C001        | North  | Laptop  |        75000 | 2023-01-10 | 2023-01-10 10:15:00 | 2026-05-08 09:51:14.728749
 18 | 1002     | C002        | South  | Mobile  |        30000 | 2023-01-12 | 2023-01-12 11:20:00 | 2026-05-08 09:51:14.728749
 19 | 1003     | C001        | North  | Tablet  |        20000 | 2023-02-05 | 2023-02-05 09:10:00 | 2026-05-08 09:51:14.728749
 20 | 1004     | C003        | East   | Laptop  |        80000 | 2023-02-20 | 2023-02-20 14:45:00 | 2026-05-08 09:51:14.728749
 21 | 1005     | C004        | West   | Mobile  |        28000 | 2023-03-01 | 2023-03-01 16:30:00 | 2026-05-08 09:51:14.728749
 22 | 1006     | C002        | South  | Laptop  |        72000 | 2023-03-15 | 2023-03-15 18:25:00 | 2026-05-08 09:51:14.728749
 23 | 1007     | C001        | North  | Mobile  |        32000 | 2023-03-18 | 2023-03-18 20:10:00 | 2026-05-08 09:51:14.728749
 24 | 1008     | C003        | East   | Tablet  |        22000 | 2023-04-02 | 2023-04-02 12:00:00 | 2026-05-08 09:51:14.728749
 25 | 1001     | C001        | North  | Laptop  |        75000 | 2023-01-10 | 2023-01-10 10:15:00 | 2026-05-08 09:51:29.58361
 26 | 1002     | C002        | South  | Mobile  |        30000 | 2023-01-12 | 2023-01-12 11:20:00 | 2026-05-08 09:51:29.58361
 27 | 1003     | C001        | North  | Tablet  |        20000 | 2023-02-05 | 2023-02-05 09:10:00 | 2026-05-08 09:51:29.58361
 28 | 1004     | C003        | East   | Laptop  |        80000 | 2023-02-20 | 2023-02-20 14:45:00 | 2026-05-08 09:51:29.58361
 29 | 1005     | C004        | West   | Mobile  |        28000 | 2023-03-01 | 2023-03-01 16:30:00 | 2026-05-08 09:51:29.58361
 30 | 1006     | C002        | South  | Laptop  |        72000 | 2023-03-15 | 2023-03-15 18:25:00 | 2026-05-08 09:51:29.58361
 31 | 1007     | C001        | North  | Mobile  |        32000 | 2023-03-18 | 2023-03-18 20:10:00 | 2026-05-08 09:51:29.58361
 32 | 1008     | C003        | East   | Tablet  |        22000 | 2023-04-02 | 2023-04-02 12:00:00 | 2026-05-08 09:51:29.58361
 33 | 1001     | C001        | North  | Laptop  |        75000 | 2023-01-10 | 2023-01-10 10:15:00 | 2026-05-08 09:53:50.506293
 34 | 1002     | C002        | South  | Mobile  |        30000 | 2023-01-12 | 2023-01-12 11:20:00 | 2026-05-08 09:53:50.506293
 35 | 1003     | C001        | North  | Tablet  |        20000 | 2023-02-05 | 2023-02-05 09:10:00 | 2026-05-08 09:53:50.506293
 36 | 1004     | C003        | East   | Laptop  |        80000 | 2023-02-20 | 2023-02-20 14:45:00 | 2026-05-08 09:53:50.506293
 37 | 1005     | C004        | West   | Mobile  |        28000 | 2023-03-01 | 2023-03-01 16:30:00 | 2026-05-08 09:53:50.506293
 38 | 1006     | C002        | South  | Laptop  |        72000 | 2023-03-15 | 2023-03-15 18:25:00 | 2026-05-08 09:53:50.506293
 39 | 1007     | C001        | North  | Mobile  |        32000 | 2023-03-18 | 2023-03-18 20:10:00 | 2026-05-08 09:53:50.506293
 40 | 1008     | C003        | East   | Tablet  |        22000 | 2023-04-02 | 2023-04-02 12:00:00 | 2026-05-08 09:53:50.506293
(40 rows)
```